2.2. SOLAR ENERGY FOR SMART CITY

India is witnessing unparalleled transformation from rural to predominantly urban living over the last two decades. It is expected that the smart cities will include a large number of infrastructure services and smart solutions. In particular, the elements that must form part of a smart city are assured electricity supply with at least 10 per cent of the smart city's energy requirement coming from solar and smart metering.

Renewable energy technologies will contribute a lot in achieving objectives of the Smart Cities Mission to make them zero polluting and self-sustaining cities. Renewable energy like Solar Energy has a lot much to contribute in that. Rooftop Solar Energy will make smart cities households self-sustaining in their energy requirements. Solar Energy is also zero polluting energy. Effective and efficient utilization of available rooftops with net metering facilities allowing excess generation to be fed into grid will definitely do the basic fulfillment of mission of self-sufficiency in green energy need.

Small solar parks can be made an integral part of Smart Cities so as to fulfill any additional such requirements. The solar energy can be harnessed in other forms like:

- rooftop solar,
- solar street lighting,
- solar water heaters,
- solar pumps,
- solar traffic signals,
- solar concentrators based cooking.

There is need for further improvement in efficiencies of Solar Production and Smart Storage Solutions so as to better achieve these objectives. Further other renewable energy technology that has to contribute more in this mission is waste-to-energy. The solid wastes that are produced can be converted into energy and other bio products that will make these smart cities also self-sustaining in energy and bio fertilizers requirements and also release zero wastes from these households.

It has already been mandated that 10 per cent of the smart cities' energy requirement will come from solar energy and at least 80 per cent buildings should be energy efficient and green buildings. Solar applications such as solar street lights, solar water heaters, rooftop solar etc, which run on Solar energy can go a long way in providing a clean and green living environment in these smart cities.

Solar energy can play a big role in creating Smart cities.

Major benefits of solar energy are:

(i) Renewable resource: Solar energy is a renewable source of energy as it can produce electricity as long as the sun exists. It is never ending source and energy can be harnessed by installing solar panels that can reduce our dependency on non- renewable sources of energy.

(ii) Eco-friendly: It is an alternative to fossil fuels as it is non-polluting, clean, reliable and renewable source of energy. It does not release harmful gases like carbon dioxide, nitrogen oxide or sulphur oxide. So, the risk of damage to the environment is reduced. Solar energy is produced from the Sun; therefore it doesn't require transportation of fuel or storage of radioactive waste.

Increased use of solar panels to produce energy will reduce the number of emissions from fossil fuels helping alleviate the damage of smog, acid rain, climate change, and contaminated water sources.

(iii) Reduced electricity bill: Interestingly the solar energy minimizes the utility bills. By installing solar panels in the houses, the cost of electricity cuts down. Solar energy can be used for activities like heating water and heating homes. This has the potential to save up to 20 per cent of the energy costs.

(iv) Easy to maintain: The cost of installing a solar panel is high, but once it is installed the benefits can be enjoyed for years with minimal maintenance. As per the needs, panels can always be added which would not require a lot of finance. These panels are noiseless and do not release any toxic substances

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(v) Installation: Solar panels can be easily installed anywhere as they do not require any wires, cords or power sources. Solar panels are mostly installed on the rooftops, therefore no new space is required for these panels.

(vi) Remote locations: Solar energy can be produced in remotest of the areas, which means it can be installed in areas where power lines would be difficult to place. Thus, helping in placement of solar panels in remotest of the areas in India.

Why Solar Power is especially suited to India

India's geographic location and conducive climate make solar energy tremendous successful in India. With over 300 sunny days through the year, the country can easily reduce dependence on fossil fuels by a significant margin given the right policy push. Currently, India ranks 5th in the global standing for solar installations. The mission, paving the path for solar in India, has set a target of 100 GW of solar power by 2021-22.

How Solar is being implemented in Smart Cities

The master plans for smart cities include both in-city and offsite power generation systems.

As for offsite infrastructure, the 100GW solar by 2022 roadmap is looking at 20 GW of power generation through solar parks. Another 10 GW will be implemented with the help of farmers.

However, the in-city mission has been given massive precedence with over 40 GW coming through rooftop solar plants. Installing solar panels on the rooftop has never been easier, thanks to advances in solar technology. The proliferation of solar PV systems across the globe has massively reduced solar panel prices as well.

Now, the usage of a rooftop solar panel for home is not just limited to water heaters. Several countries have shown massive impetus in powering entire homes and shops through PV panels. More than 16 states in the country have already set up policies to encourage solar usage. Net metering policies connect users of solar rooftop system for home to the grid and allows for passive income generation in small households.

Off-grid solar systems are also seeing an upsurge. Small commercial establishments and big homes can be entirely powered through 1KW solar panels available in the market. Off-grid solar system design includes just three main components – *an inverter, a battery and solar panels*. The installation is quite simple and reduces dependence on grid power supply significantly. This initiative is highly recommended for achieving 80% energy efficiency in buildings such as office spaces, schools, residential hubs etc.

Solar panel installation can also power streetlights, water pumps, traffic signals, road studs/blinkers and the states are being provided with the requisite resources to ensure successful implementation. Per the city, the Ministry of New and Renewable Energy is providing Rs. 50 Lakhs to implement solar. To develop solar cities, urban local bodies known as Solar City cell is being introduced. Citizens are encouraged to approach local bodies to tap into this resource and bring greener energy into everyday life.

As of 2019, India's solar installation stood at 26,984 MW on the ground while rooftop solar plants are at 1796 MW capacity. There is tremendous potential for a rapid rise in the use of rooftop solar installation as it is easy to implement and provides many economic benefits to users. A one-time investment, rooftop SPV panels are both economically and ecologically beneficial to users.

In 2018, ***Diu became the first Union Territory*** to be 100 per cent solar-powered under the smart cities mission. Through the use of solar power plants, 13 MW of solar power capacity was installed in just three years. A 9 MW solar park spread over 50 hectares is now the lifeline of the city. The 22,000 residents of Diu are enjoying the economic benefits of solar, with a 15% reduction in power tariffs. On the ecological front, success is even more worthy of celebration, as the city now saves 13,000 tonnes of carbon emissions each year.

Off-grid solar solutions can ensure continuous and uninterrupted supply of energy for commercial establishments. As an added plus, it can also serve as a passive income generator for small households.

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Therefore there is the large economic impetus to drive the use of Solar in India, and the smart cities mission is a step in the right direction. The master plan for solar in 50 cities has already been drafted by the ministry.

Urban development in India cannot occur successfully without serious consideration in terms of energy independence. In this sense, the smart cities mission paves a bright sunny future for urban residents of India, with government initiatives that wholly encourage tapping into the most abundant resource of plant earth – solar.

Aims and objectives of solar smart cities

1. Solar Cities aim to assist urban local bodies in evaluating their present energy consumption, future demand and preparing master Plans for energy savings.
2. Generation through RE installations & energy efficiency measures.
3. Each city to reduce its projected conventional energy demand by 10% over 5 years.60 cities are being developed as solar cities. Master plans for 50 solar cities prepared.
4. In solar cities, Municipal Corporations have also come forward to implement the scheme and setup RE Projects.

All Smart Cities to be taken under solar cities programme. The program includes Creation of a Renewable Energy Cell in all Smart Cities with emphasis on:

- Smart Energy
- Renewable Energy Generation
- Smart Meters
- Assured electricity supply
- Smart Waste Management
- Smart Buildings
- Sustainable Environment

The Smart Cities will have solar power generation, Solar Water Heaters, Solar PV Rooftop systems, Solar street lightings, Solar pumps, solar concentrators for steam based cooking and Solar traffic signals.

Roadmap for Solar Power by 2022

The total 100 GW solar power can be utilized in solar park, in farms by the farmers, by the Government or the private sector and Solar Rooftop. From figure 2., it is evident that the maximum power can be generated by the solar rooftop.

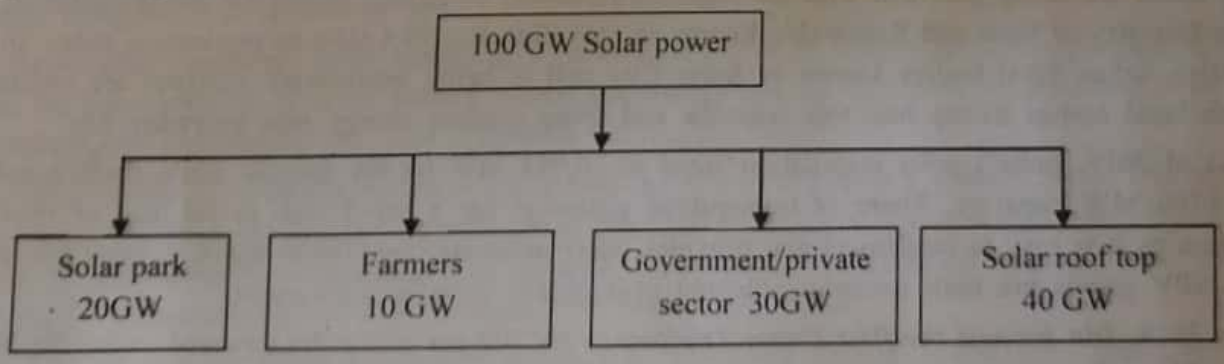

Fig. 2.1 Road map of Solar power

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The solar power installation capacity of various states in our country are given in table below.

Table 2.1 Solar power installation capacity

State		Solar power installation Capacity, MW
Rajasthan	———	1199.7
Gujarat	———	1000.65
Madhya Pradesh	———	673.58
Maharashtra	———	378.7
Andhra Pradesh	———	279.14
Tamilnadu	———	158.0

Solar radiation is the most important parameter of all the solar energy devices whether it is thermal or photovoltaic. The locations with higher radiation can be selected for installation of power plants. Rajastan, Maharastra, Gujarat and Tamilnadu states are blessed with very high radiations.

Applications of Solar Energy : Direct Solar Energy Applications are

- Solar water heating
- Space heating Space cooling
- Thermal electric conversion
- Photovoltaic electric conversion
- Solar distillation
- Solar pumping
- Solar furnace
- Solar cooking

Financial Support For Designing Smart Cities

Ministry provides up to Rs. 50 lakh per city for Preparation of a Master Plan, setting-up a solar city cell, awareness generation, to calculate capacity of building & implementation. Financial assistance can be availed for Solar systems installed on rooftops of residential, commercial, institutional & industrial buildings premises. Electricity generated could be -fed into the grid at regulated feed-in tariffs or -used for self-consumption with net-metering approach.

Issues and Challenges for solar power plants:

1. Installation of solar farms can be done by skilled and experienced people.

2. The giant solar farms have been built in desert regions to reduce the installation cost but these places are found far from cities where the power is needed. Expensive transmission lines are needed to bring the power to a distant market.

3. Maintenance costs and time can add-up since every inch of a solar panel must be kept clean to get its maximum operating efficiency.

4. The energy production only takes place when the sun is shining. Large storage systems need to be developed to provide a constant and reliable source of electricity when the radiation is not adequate in cloudy atmosphere and during night.

Solutions to overcome challenges:

1. Scientists and Engineers need to discover more efficient semiconductors or some alternative inorganic materials to increase the efficiency of the solar cell. Doubling the efficiency of solar panel will reduce the size of the array which in turn requires less space and so can shorten the payback time. Mass production of solar panels with high efficiency will bring down the costs and make them cheaper.

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2. New transmission technology is needed to bring the clean energy market. Low cost high efficient Energy storage systems can attract the users for utilization of solar energy.

3. Awareness programs are required to change the mind set of people.

Solar Vehicles:

Vehicles produces air pollution and noise pollution every day and night. If these vehicles utilize solar energy instead of conventional fuel petrol the nation will progress at faster rate as these conventional fuels are inadequate and are exported from other countries.

(a) Solar hybrid auto rickshaw:

Solar hybrid vehicles like auto rickshaws can run successfully with the help of solar electric power and can save environment by reducing air pollution and noise pollution by about 30%. In such hybrid vehicles the speed limit helps in avoiding accidents to a great extent.

(b) Solar car:

Solar car uses clean solar energy by adopting which fuel scarcity and pollution problems can be solved. Solar traffic control and traffic signals, solar warning lamps, solar marine lamps, solar air craft lamps etc. can be made.

(c) Solar power train:

India's first solar power train will be running with green initiative on Jodhpur division of North Western Railways. All the coaches of the train would be lit from power generated by the rooftop solar panels. The DEMU train which is being used for the trial run has six coaches each fitted with 12 solar panels of capacity 300W sufficient for light and fans inside the coach.

(d) Solar railway station:

A solar power plant of 1 MW capacity was set up at Katara station in 2015 which can save Rs.1Crore annually on electric bills. This is the biggest roof top solar power plant built at Jammu & Kashmir by Indian Railways.

(e) Solar boat:

Kerala Government Commissioned India's first Solar Powered boat for greener tomorrow. The solar boat runs with solar energy which is more eco-friendly and cheaper alternative to diesel. This boat has75 seats and is 20 m long. It is equipped with two electric motors running with lithium battery packs charged by a solar module of 29kWp.

(f) Solar powered rocket ships:

It is anticipated that renewable space ships will make future deliveries to space stations and possibly turn our solar system into an expensive highway or even interworking residency.

2.3 HOUSING

Towards the Smart Cities mission, the aim is to promote cities that provide core infrastructure and give a decent quality of life to its citizens. The core infrastructure elements in a smart city would include adequate water supply, sanitation, electricity, strong public transport, housing for urban poor, IT connectivity, sustainable environment, health and education, good governance and safety &security of citizens. Housing for urban poor is very important need of urban India as the population increasing rapidly in urban areas. The luxurious housing in urban areas is on the far side of affordability of majority of population belongs to middle-income and low income groups. ***Affordable housing*** is vital to have social integration and alleviating urban poverty. Simultaneously we should focus on a clean and sustainable environment while delivering the affordable housing. The provision of housing for urban poor which is sustainable in nature and should not create undue cost burden on citizens. Smart solutions are very much needed to provide more and more affordable houses in future smart cities of India.

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India is facing vast growth in construction development. As the sector growing rapidly, preserving the environment is a biggest challenge. Indian Green Building Council (IGBC) continuously works to provide tools that facilitate the adoption of green building practices in India. With the cost consideration we incorporated only five credits from the IGBC New Building Rating system which include Sustainable Architecture & Design, Site Selection and Planning, Water Conservation, Energy Efficiency and Building Materials & Resource.

⇨ **Criteria for housing in smart city:**

(1) Sustainable Architecture & Design

The housing project has to involve team members from multi-disciplinary fields for effective decision making and enhanced building performance, starting from conceptual stage to completion of project. Housing project should retain 20% of the existing topography or landscape, without any disturbance and should hand over to ULB for public use.

(2) Site Selection and Planning

All local building bylaws should be followed in the housing project. Project should get completion certificate and Environmental Clearance Certificate from respective authority. Soil erosion control measures should be taken before, during and post construction activities on project site. Site selection should be such that basic amenities like Hospital/Clinic, School, Market, Bank, Parks and Playground are within walking distance of 1 to 2 Km. Housing project located such that it should within walking distance or 1 km from one of the nearest Bus stops or any other public transport.

(3) Water Conservation

Effective rainwater harvesting enhances ground water table and reduces municipal water demand. Housing project should have on site Wastewater treatment plant to handle 100% of waste water generated in the buildings as per guidelines and reuse treated water for gardening. Housing project should demonstrate sub metering system for municipal water line and treated water consumption for better results.

(4) Energy efficiency

On site renewable energy like solar energy is used for street lighting system. From solar energy source around 2% to annual energy consumption should generated on site.

(5) Building Materials & Resource

Housing project should have separate bins to collect dry waste (Paper, Plastic, Metals, and Glass etc.) and wet waste (organic) at all the floors & common area of buildings. Use materials certified by IGBC, which are recycled and it cost at-least 10% of total building cost. These materials are manufactured locally within minimum distance which kept transportation cost low.

With rising land and construction costs, large cities must identify ways to get more from the investment in affordable housing, and to deliver housing faster to residents in need. The application of innovative construction methods, the use of alternative and sustainable materials, and new planning and cost management systems represent solutions that could facilitate more affordable housing.

Smart cities seek innovative technologies and solutions to shorten project timelines, reduce costs, and promote cleaner and more sustainable industry, to reduce the cost of construction and rehabilitation in dense urban areas and to make housing more affordable.

Solutions and technologies suited to achieve these goals could be, among others:

- Offsite Prefabrication
- Alternative Building Materials
- Building Information Modeling
- Virtual Reality / Augmented Reality
- Autonomous Drones

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- Resource Accounting and Material Flow Analysis
- 3D Printing
- Smart Devices and Mobile Based Solutions

⇨ Challenge of affordable housing

The Affordable Housing Challenge is open to the public, private, social, and academic sectors that are currently implementing innovative methods and solutions that contribute to making housing more affordable, inclusive, and efficient.

These include: city governments and municipal agents, urban developers, entrepreneurs, startups, digital innovators, developers, research centers, universities, civic techs, civil society organizations, non-governmental organizations, cooperatives, consortiums (public-private), and social activists.

⇨ Housing in cities "SWOT" analysis

Housing for low-income groups has been identified as an area of concern in almost every shortlisted Smart City Proposal though different cities have addressed the issue through varying lenses and approaches. Certain cities, including Ahmedabad, Allahabad, Bengaluru, Dehradun, Indore, Muzaffarpur, Raipur, Panaji, Patna, Rajkot, Ranchi, Sagar, Tirunelveli, and Visakhapatnam have termed 'slums,' housing for EWS/LIG, and 'illegal' housing as 'threat' in their city's 'Strengths, Weaknesses, Opportunities, and Threats (SWOT)' analysis. A few other cities termed housing for marginalized groups a 'weakness' in their analysis. Bilaspur identified growing real estate development and availability of land for redevelopment in the city as an 'opportunity' and housing shortage for EWS and the middle class as a 'weakness' in its SWOT analysis. Amaravati's proposal identifies the challenges pertaining to cohesive social inclusion due to in-migration as a 'weakness' in its SWOT analysis. Bengaluru has identified unchecked urban sprawl as a 'weakness' and encroachment on water bodies and lakes as well as the growing population with no access to 'affordable' housing and basic amenities in the city as 'threats.' The cities of Chandigarh and Dharamshala termed their approach to housing provisioning as 'strengths.' While Chandigarh claims to have already handed over housing units to half the households it identified for rehabilitation, only 0.5 per cent of Dharamshala's population is claimed to be houseless.

Bhopal's proposal noted that 75 per cent of the city's population belongs to EWS/LIG while Warangal reported EWS/LIG as accounting for 42 per cent of its population. Ahmedabad, in its proposal, reported 13 per cent of all households in the city (as of 2011) living in substandard housing while Karimnagar stated that 29 per cent of its population lives in substandard housing without basic facilities.

2.4 SUSTAINABLE GREEN BUILDING (June-2022)

Our ancestors worshipped the five elements of nature -

Earth as **'Prithvi'**

Water as **'Jal'**

Agni as **'Energy'**

Air as **'Vayu'**

Sky as **'Akash'.**

A **green building** or **environmental building** or **sustainable building** is a building which seek to harness all these natural elements in its design and construction to an optimum level so as to have an eco-friendly, low energy & low water consumption building and at the same time providing fresh and healthy environment to its occupants.

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The term 'green building' apply not just to products, but building design, orientation, construction strategies, landscaping, building operations, maintenance, and more. The less impact a building has on human health and the environment, the more green it is.

The concept of green building incorporates and integrates a variety of strategies during its planning, design, construction and maintenance of building projects. Common goal of green building is achieving aesthetic harmony between a structure and its surrounding environment and use those resources that help in environmental protection or disturbs least to the environment compared to other alternatives available. Green building movement is infact the result of contribution from architects and engineers in maintaining the ecological balance, thus, a step forward by architects and engineers to work as an environmentalist.

As regards the cost impact in the design of a green building, the cost could be slightly higher than a conventional building. However green buildings, on account of their design features, consume less energy and less water vis-a-vis a conventional building. Thus this extra cost gets paid back in 3-5 years time.

- **Objectives of Green Building :**

The design, construction and maintenance of buildings have a tremendous impact on our environment and our natural resources. Traditional building practices often overlook the interrelationships between a building, its components, its surroundings, and its occupants; and consume more of our resources than necessary, negatively impacting the environment. The basic resources like ground cover, water, forests and energy are depleted to construct and operate buildings. This is where green buildings step in with an aim to address all these issues in an integrated and scientific manner.

The main objective of green building is to reduce the overall impact of building on environment and on human health by minimising the usage of natural resources (like ground cover, water, forests, energy) and to maximize the benefits.

1. To minimise the use of natural resources like ground cover, water, forests, energy, etc.
2. To reduce overall impact of building on environment and human health.
3. To reduce waste, pollution and environmental degradation.
4. To select proper building materials and use recycled and scrap materials having low energy requirement.
5. To boost renewable energy usage, i.e. solar energy and wind energy.
6. To enhance interior lighting quality by increased use of natural day lighting to reduce energy demand.
7. To make efficient use of water in a building, reduce wastage and water pollution.
8. To improve human health and comfort.
9. To reduce global warming.

2.4.1 FEATURES (PRINCIPLES) OF GREEN BUILDING (Dec.-2021)

The different elements of a green building are as follows :

1. Sustainable site selection
2. Energy efficiency
3. Water efficiency
4. Materials efficiency
5. Design efficiency
6. Occupant Health and safety

1. Sustainable site selection :

Site selection, its development and construction process are basic considerations for any project. The impact of project should be minimum on the ecology, so that the nature adjusts itself within a short period.

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Therefore, the first objective of site selection is to select a site that does not disturb ecological balance of the area. The natural landscaping and natural features not be destructed as far as possible.

2. Energy efficiency : **[Jan. 2019]**

Building consumes energy mainly for cooling, heating and lighting. The design of a building should be such that it can use maximum available energy of the nature and this type of design is termed as **passive design of a building.** Passive design of a building can dramatically change the energy use pattern of a building. The measures include building shape and orientation, passive solar design, and the use of natural lighting.

Some of the energy saving measures are :

- Use a properly sized and energy efficient heating /cooling system in a building.
- Use solar energy for lighting, water heating.
- Maximum use of natural lighting and air.
- Maximize light colours for roofing and wall finishing materials.
- Use minimam glass on east and west exposures.
- Insulate the walls. The options for insulation materials can be - Extruded polystyrene, Expanded polystyrene (thermocol), glass wool, etc.
- Brick wall with air cavity can also significantly reduce the heat ingress.
- Hollow blocks, fly ash bricks, Autoclaved Aerated concrete (AAC) blocks are also good insulators.
- Bureau of Energy efficiency (BEE) star rated lamps and tubelights should be used.

3. Water efficiency :

Water is going to be a scare natural commodity, leading to clashes between users and even between governments. Thus neither it can be wasted nor it can be polluted. Thus water efficiency can be achieved only if water is efficiently used without wastage, reused to the maximum extent and water pollution is stopped.

Avoiding wastage is considered better than production. To achieve this an integrated and holistic approach is to be adopted for water management.

- **Measures to improve water efficiency :**
  - Potable water should not be used for the purposes other than essential requirements.
  - For other uses waste water is to be recycled. Waste water treatment plant should be installed at the project site.
  - In buildings water saving materials are to be used like low capacity cisterns, showers, waterless urinals, etc.
  - Water meters should be installed.
  - Use waste water for floor washing and gardening.
  - Instead of continuous supply of water adopt intermittent system of supply.

4. Materials efficiency :

Selection of new construction materials and use of recycled and scrap materials are the two important factors for the construction of green buildings. Such products promote resource conservation and efficiency. Using recycled materials also helps in solid waste management system.

Building materials consume a considerable energy and affect ecological balance due to extraction from

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the nature. Building material is to be examined for being a green material based on its energy requirement from raw stage to the providing and fixing stage in a building. These stages may be extraction, manufacturing, transportation, fixing, maintenance and disposal.

Green building materials offer the following benefits :

- Energy conservation
- Lower costs
- Greater design flexibility
- Reduced maintenance costs
- Improved occupant health and productivity

Various Green Materials :

(i) Fly ash

(ii) Green Concrete

(iii) Use of scrap and recycled aggregate in concrete.

(iv) Other waste materials such as

- Blast furance slag
- Silica fume
- red mud
- Marble dust
- cinder
- rice husk
- coconut husk
- waste glass
- Jute fibres
- rubber from automobile tires
- Bamboo strips

have also been demonstrated by research.

(v) Wood salvaged from old buildings has been used for doors and windows.

5. Design efficiency :

Design of a building is very important as it plays a vital role in reducing the energy requirement of a building during its life.

Design has to consider the following :

(i) Optimum use of solar energy.

(ii) Optimum use of renewable energy.

(iii) Selection of energy efficient fittings, plants and equipment.

(iv) Adoption of environment friendly and energy efficient materials.

(v) Adopting energy efficient design and technologies.

A good architectural design will take full advantage of sun, wind and trees around the building. The location of door, windows and walls will be designed in a way to have maximum advantage of nature to minimize energy requirements. While deciding the specifications of materials and adoption of technologies for execution of the work, civil engineer plays an important role.

6. Occupant health and safety :

Research findings reported in different literatures revealed that buildings with good overall environmental quality can reduce the rate of respiratory disease, allergy, asthma, and enhance worker performance. Choose construction materials and interior finish products with zero or low emissions to improve indoor air quality. Many building materials and cleaning/maintenance products emit toxic gases, such as volatile organic

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compounds (voc) and formaldehyde. These gases can have a detrimental impact on occupant's health and productivity .

Provide adequate ventilation and a high-efficiency, in-duct filtration system. Heating and cooling systems that ensure adequate ventilation and proper filtration can have a dramatic and positive impact on indoor air quality. Provide effective drainage from the roof and surrounding landscape.

2.4.2 BENEFITS OF GREEN BUILDING

(a) Economic benefit (tangible benefits) : **(June-2022)**

1. **Energy efficiency :** Green buildings are around 25-30% more energy efficient, with gold-rated buildings as much as 37% efficient.
2. **Water efficiency :** Green buildings use 20-30% less water compared with similar conventional buildings. This reduces operating water costs.
3. **Waste reduction :** Green buildings emphasise waste reduction. Construction wastes and demolition debris are the main wastes produced during the construction process, which degrade the quality of environment.
   - The reuse and minimisation of construction wastes and debris and diverting them to recycling units.
   - The increased use of recycled content in construction materials.
4. Improve employee productivity and satisfaction.
5. Enhance asset value and profits.
6. Optimize life-cycle economic performance.
7. Incorporate latest techniques and technologies.

(b) Environmental benefits : **(Dec.-2021)**

1. Enhance and protect ecosystems and biodiversity.
2. Improve air and water quality
3. Reduce solid waste
4. Conserve natural resources
5. Reduced water consumption

(c) Health and community benefits :

1. Improve air, thermal and acoustic environment.
2. Enhance occupant comfort and health.
3. Contribute to overall quality of life.
4. Enhanced image and marketability

2.5 SAFETY AND SECURITY **(Dec.-2021)**

These days we see more and more people migrating to urban areas to create a better life for themselves. This, of course, places a greater urgency on governments and metropolis to improve public safety and security. However, we do know that new technologies and devices enable urban areas to deploy smarter applications for crime detection and prevention. These applications are crucial to helping the police, medical teams and first responders make the best decision possible when dealing with emergency situations.

Given the critical importance of security in cities, innovations in the IoT (Internet of Things) are increasingly improving the safety of city inhabitants. New services such as remotely connected CCTV and automated incident detection allow a quicker response to threats. IoT led crowd management solutions help protect city inhabitants in busy, crowded areas. Smart street lighting, which helps to lower crime rates and improve driving conditions, is another example of the measures city planners are taking to improve people's safety.

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